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**Title:** Antibiotic resistance and random amplified polymorphic DNA typing of *Klebsiella pneumoniae* isolated from clinical and water samples

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/wer.1630

**Title:** Antibiotic resistance and random amplified polymorphic DNA typing of *Klebsiella pneumoniae* isolated from clinical and water samples


## Abstract

The study aimed to screen for the presence of multidrug-resistant *Klebsiella pneumoniae* from diarrheal stool and environmental water samples and to check the epidemiological link between the two categories. Isolates obtained after culturing on different media were tested for antibiotic resistance and extended-spectrum beta-lactamase (ESBL) production. Polymerase chain reaction (PCR) analysis was done for important β-lactamase encoding genes. Random amplified polymorphic DNA (RAPD) typing was done using two primers. Results showed a high prevalence of *K. pneumoniae* from fish market effluents compared to stool and well water. Stool isolates showed high resistance to ceftazidime (80.0%) and cefepime (80.0%), fish market effluent isolates to cefoperazone-sulbactam (92.1%), and erythromycin (78.9%), while well water isolates to erythromycin (72.7%) and cefuroxime (54.4%). The ESBL genes $bla_{\mathrm{CTX}}$, $bla_{\mathrm{SHV}}$ and $bla_{\mathrm{TEM}}$ were detected in 22.85%, 14.28% and 42.85% of *K. pneumoniae* isolates, respectively. The results of RAPD-PCR showed high genetic similarities between the isolates from different sources.



**Keywords:** *Klebsiella pneumoniae*, microbial contamination, wastewater, antibiotics, environmental health, PCR, RAPD

## Introduction

*Klebsiella* species are the primary cause of healthcare and community-acquired infections associated with 10% of nosocomial infections, particularly those involving the urinary, respiratory and gastrointestinal tracts [Chong et al. 2018]. Studies on epidemiological investigation have suggested that majority of *K. pneumoniae* infections are preceded by intestinal colonization [Lin et al. 2012; Martin et al. 2016]. A study from Japan described the spread of a virulent *K. pneumoniae* clone among the family members, showing evidence that the virulent clones colonized the intestinal tracts of the family members for at least two years [Harada et al. 2011]. Of additional concern, *K. pneumoniae* acquire, accumulate, and transfer antimicrobial resistance determinants and therefore may represent a significant reservoir for resistance within the gut [Huddleston 2014], and may increase the risk of resistant infections in hospital environments [Taitt et al. 2017].

*K. pneumoniae* is known for its resistance properties to most of the last-line antibiotics that are usually used. The resistance to third-generation cephalosporins, carbapenems, and fluoroquinolones was first described in the 1980s. Since then, the prevalence of ESBL-producing *Klebsiella* in hospitals has progressively increased [Saadatian et al. 2018]. India records a high prevalence of antibiotic-resistant *Klebsiella* with reported frequencies of 14-100% [Fouzia and Damle 2015]. Antibiotic-resistant *K. pneumoniae* has been isolated from a variety of retail meats, seafood, and stagnant water [Runcharoen et al. 2017], with several of these environmental isolates having a high epidemiological link to clinical *K. pneumoniae* strains [Runcharoen et al. 2017; Chi et al. 2019].

Infections caused by ESBL-producing organisms are difficult to treat since the therapeutic antibiotic choices are limited. ESBLs hydrolyze third-generation cephalosporins and aztreonam, but not carbapenems, and are inhibited by clavulanic acid and tazobactam. However, the emergence of carbapenem-resistant *K. pneumoniae* is a significant public health concern given minimal antibiotic treatment options [Haverkate et al. 2016]. Several types of ESBLs are prevalent, of which ESBL- encoding genes *bla*$_{TEM}$, *bla*$_{CTX}$, and *bla*$_{SHV}$ are of significant importance [Sanjit et al. 2017]. An Indian study has reported β-lactamase genes from untreated hospital wastewater effluents [Walsh et al. 2011]. The research suggests that contaminated fish and its effluents can be sources of environmental contamination if not treated effectively.

Several PCR-based DNA fingerprinting methods are available for epidemiological differentiation of *Klebsiella* isolates, such as pulsed-field gel electrophoresis (PFGE), restriction fragment length polymorphism (RFLP), plasmid-profile analysis, ribotyping, and random amplified polymorphic DNA. RAPD is a PCR-based genotyping technique that targets random segments of the genomic DNA using a single primer of arbitrary nucleotide sequence [Williams et al. 1990]. It is a rapid, simple, and inexpensive method to determine the clonal relatedness among bacteria isolated from varied sources. This technique has been used widely for *Klebsiella sp*. for diversity studies [Ben-Hamouda et al. 2003]. Most previous studies have focused on strains isolated from clinical respiratory specimens, and little is known about distribution and characteristics of *K. pneumoniae* in faeces of diarrhea patients. Since minimal data is available regarding the molecular epidemiology of ESBL-producing *Klebsiella*, this study aimed to isolate *K. pneumoniae* from diarrheal stool and environmental sources and further screen the isolates for antibiotic resistance and ESBL production to evaluate the burden of fecal carriage of resistant isolates disseminating into the environment. The *K. pneumoniae* strains were also genotyped by RAPD-PCR to look into the epidemiological link between the clinical and environmental isolates.

## Materials and Methods

### *Sample collection*

In this study, 150 samples were collected between February-December 2018 and analyzed for *K. pneumoniae.* The sample source included clinical samples (n=50), well water (n=50), and fish market effluents (n=50). Clinical samples comprised of diarrheal stool collected from patients attending to a tertiary care hospital in Mangaluru, India. Environmental samples were obtained within a 15 km range from the hospital to compare outcomes.

A demographic survey undertaken to understand the source of infection indicated that the hospitalized patients resided within a 15 km radius of the hospital and close to fish markets. Hence, in this study, the environmental samples included well water collected from wells near the patient's residence, which was used as a drinking water source. Well water samples were further classified as residential, market, and hospital, depending on the collection area. Since our survey also indicated the possibility of the let-out fish market effluents seep and cross-contaminate the nearby well waters, we also undertook to analyze the fish market effluents for *K. pneumoniae* in this study. Fish market effluent samples were collected from various effluent discharge sites prior to release into the sewage from 15 different fish markets on a monthly basis. Hence, four different samples (once every month) were collected from different sites of the same fish market.

The study protocol was approved by the Central ethics committee (Ref: NU/CEC/2018/0175, dated 19/01/2018). Written informed consent was obtained from all the participants for collecting diarrheal stool. Immunocompromised patients, new-borns, and patients with chronic debilitating illnesses were excluded from the study.

### *Isolation and Identification of Klebsiella pneumoniae*

*Diarrheal stool samples:*

Approximately 5 mL was collected in a clean, dry container with a tight lid (HiMedia, Mumbai, India) labelled with patient details and transported to the Microbiology lab within two hours of collection. The sample was directly inoculated onto a basic agar i.e., Nutrient agar (NA) and a differential agar i.e., MacConkey agar (MA) (HiMedia, Mumbai, India) plate, and a part of it was inoculated into phosphate buffer saline (PBS) (HiMedia, Mumbai, India) and incubated for homogenizing and enrichment. Subculturing was done by streaking from PBS onto MA and incubated at 37ºC for 24-48 hours [Shetty et al. 2012].

*Environmental samples:*

Approximately 250 mL of each sample was collected in a sterile plastic container and transported immediately to the laboratory in a cooler with ice packs (< 4°C) and processed within 4-6 hours. For isolating the colonies, the samples were serially diluted ($10^{-1}$, $10^{-2}$, $10^{-3}$) with sterile physiological saline and 100 µl from each aliquot was spread plated onto tryptic soy agar (TSA) (HiMedia, Mumbai, India) and MA plates followed by incubation at 37ºC for 24 hours [Ibrahim and Hameed 2015]. Pink mucoid colonies from MacConkey agar were picked up and subcultured onto NA and incubated at 37ºC for 24 hours, following which the isolates were subjected to a battery of biochemical tests for identification [MacFaddin 2000]. Confirmed *K. pneumoniae* isolates were preserved in Luria Bertani (LB) broth (HiMedia, Mumbai, India) with 30% glycerol and stored at -20°C until further use.

### *Antibiogram analysis*

Modified Kirby-Bauer disk diffusion technique [Bauer 1966] was used to detect the isolates' antibiotic susceptibility inoculated on Muller-Hinton agar (MHA) (HiMedia, Mumbai, India). The antibiotic discs (HiMedia, Mumbai, India) included were: ampicillin (30 µg), amikacin

(30 µg), amoxyclav (20/10 µg), ceftazidime (30 µg), ceftazidime clavulanic acid (30/10 µg), ceftriaxone (30 µg), cefotaxime (30 µg), cefuroxime (30 µg), cefepime (30 µg), cefoperazone-sulbactam (75/15 µg), co-trimoxazole (25 µg), chloramphenicol (30 µg), ciprofloxacin (5 µg), aztreonam (30 µg), imipenem (10 µg), meropenem (10 µg), ertapenem (10 µg), gentamicin (10 µg), tigecycline (15 µg), tetracycline (30 µg) and piperacillin-tazobactam (100/10 µg). The inhibition zones were measured and interpreted as sensitive or resistant as recommended by the Clinical and Laboratory Standards Institute [CLSI 2017] guidelines using ATCC (USA) standard culture of *K. pneumoniae* 76003 as positive and *Escherichia coli* 25922 as the negative control. The multiple antibiotic resistance (MAR) index was determined as the ratio of the total antibiotics used to the number of antibiotics to which the bacterial isolate was resistant [Krumperman 1983].

***Phenotypic detection of ESBL***

The isolates were initially screened for ESBL production by double disk diffusion test [CLSI 2014] using ceftazidime (30 µg), cefotaxime (30 µg), and ceftriaxone (30 µg) on inoculated MHA plates followed by incubation at 37°C for 24 hours. Strains were considered potential ESBL producers when the inhibition zone was ≤ 27 mm for cefotaxime, ≤ 22 mm for ceftazidime, and ≤ 25 mm for ceftriaxone. Further confirmation was done using cefotaxime (30 µg) and ceftazidime (30 µg) alone and in combination with inhibitor clavulanic acid (30/10 µg). An isolate was phenotypically confirmed as an ESBL producer when the difference in the zone of inhibition of the drug and the inhibitor was ≥ 5 mm compared to the cephalosporin alone [CLSI 2017].

***DNA extraction***

Bacterial DNA was extracted using the Cetyl trimethyl ammonium bromide (CTAB) method [Ausubel et al. 1995]. Briefly, the bacterial culture grown overnight at 37$^o$C in Luria Bertani broth was centrifuged at 11,200 g for 5 minutes. The bacterial pellet obtained was resuspended in 567 µl of 1× TE buffer (10 Mm Tris-HCL, 1 Mm EDTA; pH: 8.0), 30 µl of 10 % SDS and 3 µl of 20 mg mL$^{-1}$ proteinase K. The mixture was incubated at 37°C for 1 hour following which, 100 µl of 5M NaCl and 80 µl of CTAB/ NaCl solution was added, mixed thoroughly and incubated for 10 minutes at 65°C in a water bath. An equal volume of chloroform/ isoamyl alcohol mixture (24:1) was added, vortexed, and centrifuged at 11,200 g for 5 minutes. The aqueous phase was transferred to a fresh tube and extracted by adding an equal volume of phenol/ chloroform/ isoamyl alcohol (P: C: IAA in 25:24:1) mixture and by centrifuging at 11,200 g for 10 minutes. The supernatant was transferred to a fresh tube, and DNA was precipitated by adding 0.6 volumes of isopropanol followed by centrifugation at 11,200 g for 5 minutes. DNA pellet was washed with 1 mL of 70% alcohol by centrifugation at 11,200 g for 5 minutes and was vacuum dried. Finally, the DNA was dissolved in 100 µl of 1× TE buffer and stored at -20°C for further use. The purity and DNA concentration was estimated using a Nanodrop spectrophotometer (ND-1000, V3.3.0, Thermo Fisher Scientific, USA)

***Detection of ESBL genes by PCR***

PCR assay was performed to detect β-lactamase genes *bla*$_{\mathrm{CTX}}$, *bla*$_{\mathrm{SHV}}$, and *bla*$_{\mathrm{TEM}}$ (Table 1). PCR amplification was carried out in a 30 µl reaction mixture containing 22.2 µl of sterile distilled water, 0.2 µl Taq DNA polymerase, 3 µl of Taq buffer with MgCl2 (TaKaRa, Bangalore, India), 0.6 µl dNTP, 1 µl of 10 pM primer (each forward and reverse) and 2 µl of template DNA. The primer sequences were obtained from Juniper Life Sciences, Bangalore, India. Primers thermal cycling program, their expected amplicon sizes and references are given in (Table 1). Amplification was carried out in an Eppendorf Mastercycler nexus GX2 (Thermo Fisher Scientific, Waltham, MA, USA) with the optimized PCR program. PCR products (10 μl) were analyzed by electrophoresis in 1.5% agarose gels (HiMedia, Mumbai, India) and visualized with a gel documentation system (Bio-Rad, CA, USA).

***Random amplified polymorphic DNA (RAPD)***

Two 10-mer oligonucleotide primers, 640 and 650 (Juniper Life Sciences, Bangalore, India) were initially screened to determine their optimal annealing temperature and the consistency to generate reproducible bands in the RAPD-PCR assay [Haryani et al. 2007; Ashayeri et al. 2012]. The primer annealing temperature was determined using a temperature gradient of 34°C to 42°C. The optimal annealing temperature obtained was 38.0°C for 2 minutes.

RAPD was carried out using both the primers respectively in a 30 μl reaction mixture consisting of 3 μl of 10X PCR buffer (100 Mm Tris-CL pH 8.3, 20 Mm MgCl2, 500 mM KCL, 0.1% gelatin), 250 mM of each deoxynucleotide triphosphates, 50 pM of primer and 1.5 U of Taq polymerase. Amplification was carried out in a thermocycler (MJ Research, Watertown, USA) using optimum cycling conditions: initial denaturation at 95°C for 5 minutes, 35 cycles of 1 min at 94°C, annealing temperature of 38°C for 2 min, 2 min at 72°C and a final extension at 72°C for 10 min. The amplified products were resolved in 1.5% agarose gel and stained with ethidium bromide. The gels were visualized by the Gel Documentation system (Biorad, CA, USA).

The amplicons generated in the RAPD-PCR assay for the two primers used were analyzed separately. Only samples exhibiting RAPD profiles of three or more bands were considered for cluster analysis. The RAPD patterns generated for different gels were analyzed using Gel Compare II version 2.5 (Applied Maths, St Martens-Latem, Belgium). Cluster analysis was performed based on Pearson's correlation coefficient and grouped using the unweighted pair group method with arithmetic mean [UPGMA]. The similarity levels between the profiles generated were expressed as percentages and presented as a dendrogram. The numerical discriminatory index value for the ability of the isolates to be discriminated using typing methods was calculated as per Hunter and Gaston's procedure [Hunter and Gaston 1988].

***Statistical analysis***

The data were analyzed using Statistical Package for social sciences software version 16 (Chicago, SPSS Inc). Descriptive statistics were performed. Antibiotic resistance data were treated as a binary variable (1= susceptible; 2= resistant). The collected information was summarized using frequency and percentage for qualitative data. To find the statistical association between antibiotic susceptibility and *K. pneumoniae* isolated from clinical and environmental sources, Fisher's exact test was used. *P*-value $< 0.05$ was considered statistically significant (Table 2).

## Results

***Prevalence of ESBL and non ESBL-producing Klebsiella pneumoniae***

Samples (n=150), which included both clinical (n=50) and environmental (n=100), were screened for the presence of *K. pneumoniae*. A single isolate of *K. pneumoniae* was considered from a sample to ensure non-repetitive isolates. Sixty-four samples confirmed positive for *K. pneumoniae,* of which 38 (59.37%) were from fish market effluents, 11 (17.18%) from well water, and 15 (23.43%) from diarrheal stool samples. All the 64 *K. pneumoniae* isolates were tested for phenotypic detection of ESBL. Of 64 isolates, 16 (25%) were phenotypically positive for ESBL. Of these, four (26.67%) isolates were from 15 stool *K. pneumoniae* isolates and 12 (31.57%) from 38 fish market *K. pneumoniae* isolates. Well water was negative for ESBL by phenotypic method.

***Antibiotic resistance profile***

*K. pneumoniae* isolated from stool (n=15) showed high resistance to ampicillin (86.6%) followed by ceftazidime (80%), cefepime (80%) and ceftazidime clavulanic acid (66.6%). Isolates from fish market effluents (n=38) showed high resistance to ampicillin (92.1%), cefoperazone-sulbactam (92.1%), and piperacillin-tazobactam (42.1%). Well water isolates (n=11) showed high resistance to ampicillin (100%), cefuroxime (54.5%), ertapenem (36.3%), and meropenem (36%). *K. pneumoniae* isolates from well water were negative for phenotypic detection of ESBL, but showed a significant antibiotic resistance to carbapenems (ertapenem and meropenem). In this study, majority of the isolates were highly resistant to third and fourth-generation cephalosporins and β-lactams (Figure 1). A significant association ($p$ <0.05) existed between the antibiotic susceptibility of *K. pneumoniae* isolates with different sample types like stool, fish market effluents and well water (Table 2). The antibiotic profile and the MAR index calculated are given in Table 3.

***PCR for detection of bla$_{CTX}$ , bla$_{SHV}$ and bla$_{TEM}$ genes***

The expected band sizes for *bla*$_{\mathrm{CTX}}$, *bla*$_{\mathrm{SHV}}$, and *bla*$_{\mathrm{TEM}}$ were observed at 569 bp, 1018 bp, and 356 bp, respectively (Figure 2). Molecular detection of β-lactamase genes in this study highlights the high prevalence of gene *bla*$_{\mathrm{TEM}}$, followed by *bla*$_{\mathrm{CTX}}$ and *bla*$_{\mathrm{SHV}}$. The presence of more than two genes was noted in 25.71% of isolates. There were 7 (20%) for both *bla*$_{\mathrm{CTX}}$ and *bla*$_{\mathrm{TEM}}$, 2 (5.71%) isolates positive for both *bla*$_{\mathrm{CTX}}$ and *bla*$_{\mathrm{SHV}}$, and 2 (5.71%) were positive for *bla*$_{\mathrm{SHV}}$ and *bla*$_{\mathrm{TEM}}$ genes. *bla*$_{\mathrm{TEM}}$ gene alone was possessed by 42.85% of isolates, 22.85% had the *bla*$_{\mathrm{CTX}}$ gene alone, and 14.28% isolates were positive for *bla*$_{\mathrm{SHV}}$ alone. All the three genes were positive for only 1 (2.85%) isolate, a diarrheal stool sample.

***Random amplified polymorphic DNA (RAPD)***

All the isolates were typeable using primers 640 and 650 and produced reproducible RAPD fingerprints. RAPD-PCR was run thrice to check for the reproducibility of the fingerprints generated. However, in comparison to primer 650, primer 640 resolved better and showed consistency in generating DNA band patterns. RAPD-PCR fingerprints were generated for 35 *K. pneumoniae* isolates using primer 640, which amplified 6-12 bands with sizes ranging from 0.2 to 3 Kb. A dendrogram generated based on the similarity matrix values showed heterogeneity among the *Klebsiella* isolates. At a 60% average similarity value, the strains clustered into two major groups A and B (Figure 3). Cluster A grouped 23 isolates (69.7%) comprising clinical, residential well water, and market well water samples.

On the other hand, Cluster B grouped exclusively fish market effluent isolates and one market well water isolate. The Discriminatory index (DI) of strains was achieved at 93% similarity index (DI=0.90). Cluster A comprising of 23 strains, was further sub-grouped into three clusters, A1 ( 2 strains), A2 ( 8 strains), and A3 (2 strains), while the remaining were all singletons. Similarly, Cluster B had two subgroups, B1 with eight strains and B2 with two strains (Figure 3).

***RAPD similarities with cluster-based antibiotic resistance patterns***

Table-4 shows the comparison of the isolates belonging to the same RAPD cluster and their antibiotic resistance profiles. Cluster A1 comprising two isolates, KP14 (clinical) and KP21 (hospital well water), showed similarity in terms of resistance to ampicillin, chloramphenicol and cefuroxime. Clinical and different well-water isolates of Cluster A2 also showed a similar resistance pattern towards cephalosporins and carbapenems, but the highest resistance was observed for cefepime, cefuroxime and ceftazidime-clavulanic acid. The resistance patterns of cluster A1 and A2 isolates points towards the possible cross-contamination among the well water and the clinical isolates. Two residential well water isolates falling into Cluster A3 showed similar resistance to ampicillin and erythromycin. Overall, genetic relatedness was observed among the sub-clusters of Cluster-A with resistance to imipenem, meropenem, ertapenem, ampicillin, cefuroxime and gentamicin.

On the other hand, eight fingerprints of Cluster-B1 (sub-cluster of Cluster-B) showed similarity in resistance to ceftazidime, ceftazidime-clavulanic acid, cefoperazone-sulbactam and piperacillin-tazobactam. Two fish-market effluent isolates of sub-cluster B2 were resistant to ampicillin, ceftazidime, cefuroxime, cefoperazone-sulbactam, erythromycin and piperacillin-tazobactam. Both the sub-clusters B1 and B2 showed similar fingerprints, being resistant to ceftazidime, ceftazidime-clavulanic acid, cefotaxime, cefepime, cefoperazone-sulbactam, meropenem, erythromycin and piperacillin-tazobactam apart from the intrinsic resistance to ampicillin.

***Multidrug resistance in Klebsiella pneumoniae isolates***

Of the 64 isolates, 62 (96.87%) were resistant to two or more antibiotics tested with a MAR index of > 0.09, and there were 54 (84.37%) isolates that were resistant to four or more antibiotics with a MAR index of > 0.19 (Table 3). Out of the total 21 antibiotics tested, two stool isolates were resistant to 17 (80.95%) antibiotics, and one stool isolate was resistant to 16 (76.19%) antibiotics. One fish market effluent strain was resistant to 15 (71.42%) out of 21 antibiotics tested. The majority of *Klebsiella* isolates from this study belong to the > 0.2 MAR index.

**Discussion**

From the last few years, the rapid increase in the incidence of ESBL-producing *Enterobacteriaceae* in hospital settings and subsequent spread to the community has become a worldwide concern for public health. A study from the Netherlands showed that although human health care settings were initially responsible for ESBL-producing organisms, they could now be easily isolated from several food and animal sources [Van et al. 2015]. In this study, 25 % of the *K. pneumoniae* strains were found to be ESBL positive. This is in accordance with earlier studies from India where the prevalence was recorded to be between 17-70% [Sharma et al. 2013; Sarojamma and Ramakrishna. 2011; Shanthi and Sekar. 2010].

This study is a prospective, observational work done on mainly two groups, i.e., clinical and environmental isolates. Two different environmental samples were collected to analyze the multidrug-resistance pattern disseminating from the humans to the ecosystem and vice versa. Our observations highlight a similar pattern of resistance for both groups. The majority of *K. pneumoniae* isolates from this study showed MAR index >0.2. A MAR index of >0.2 suggests an extensive use of antibiotics and contamination from high-risk sources [Krumperman 1983]. Thus, the *K. pneumoniae* isolates exhibiting high resistance to major antibiotic classes demonstrate the associated human health risk when entering the food chain and becoming passive carriers. In similar lines, a study from Maryland highlighted reports of patient-to-patient transmission in 52% patients who acquired colonization with ESBL-producing *K. pneumoniae* [Harris et al. 2007].

Some of the isolates from stool and fish market effluents, that were resistant to β-lactam drugs such as imipenem, meropenem, cefepime, cefotaxime and ampicillin, also showed resistance to β-lactamase inhibitors cefoperazone-sulbactam and piperacillin-tazobactam. β-lactamase inhibitors are known to prevent bacterial degradation of β-lactam antibiotics [Wright 2005], and resistance to these inhibitors will have a significant impact on the treatment of infectious diseases. In this study, well water isolates were 100% sensitive to quinolones and showed least resistance to aminoglycosides (100% sensitivity to amikacin and 9% resistance to gentamicin), but 46.6% of stool isolates and 7.8% of fish market effluents were resistant to ciprofloxacin. Also, 46.6% and 6.6% of stool isolates were resistant to gentamicin and amikacin, respectively and 13.1% and 10.5% of fish market effluents were resistant to gentamicin and amikacin, respectively. Hence, this study highlights significant resistance to carbapenems, quinolones and aminoglycosides, mainly in stool and fish market effluent isolates.

The emergence of plasmid-mediated ESBLs among *Enterobacteriaceae* members has increased worldwide [Rodrigues et al. 2004; Ashayeri et al. 2012]. In agreement with our observations, a study from Tunisia reports the detection of *K. pneumoniae* isolated from different hospital environments, which showed high resistance to cephalosporins and carbapenems, suggesting the circulation of ESBL and carbapenemase encoding genes in the environment [Dziri et al. 2016]. Our study shows a high prevalence of $bla_{\mathrm{TEM}}$ gene (42.85%) among *K. pneumoniae*, followed by $bla_{\mathrm{CTX}}$ (22.85%) and $bla_{\mathrm{SHV}}$ (14.28%). Similar research from South Korea reports $bla_{\mathrm{TEM}}$ and $bla_{\mathrm{SHV}}$ gene in abundance from all the isolates of fish-farm effluent samples but not $bla_{\mathrm{CTX}}$. The high prevalence of these genes from fish aquaculture sites situated near the marine environment clearly shows an increased usage of beta-lactam antibiotics in fish farming [Jang et al. 2018], which eventually contaminates the seafood. There are high chances of these resistant genotypes circulating in the community, indicating an increased local antibiotic consumption pattern.

Moreover, from this study, isolates (KP25, KP8, KP10 and KP11) that were negative for phenotypic detection of ESBL and susceptible to β-lactams and lactam-inhibitor combinations, detected *bla*-genes genotypically ($bla_{\mathrm{CTX}}^{+}$ and $bla_{\mathrm{TEM}}^{+}$). These discrepancies could be due to poor gene expression or non-functionality of the expressed gene products. All the phenotypically positive ESBL isolates showed the presence of at least one *bla*-gene by PCR.

Molecular typing method's efficiency can be assessed by factors such as typability, reproducibility, and discriminatory power. The fingerprinting of *K. pneumoniae* isolates demonstrated that the strains were genetically diverse and heterogeneous. No relationship was observed between the specimen type or ESBL production with the isolates clustered (Figure 3). Among the two primers, Primer 640 produced well distinguishable and reproducible amplicons showing better clustering compared to primer 650. Two significant clusters were obtained at a discriminatory index of 0.90 using primer 640 whereas, primer 650 produced two significant clusters and the rest all singletons at a similarity rate of 90% and DI= 0.94. According to Hunter and Gaston, a discriminatory index value greater than 0.90 is desirable to confidently interpret the discrimination levels for RAPD typing [Hunter and Gaston 1988], which agrees with our observations where primer 640 offered sufficient discriminatory power (≥ 0.90). Most of the isolates belonged to cluster A. However, the isolate's antibiogram patterns were different from each other, certain degree of similarity existed between the sub-cluster fingerprints and their antibiotic resistance pattern (Table 4). The presence of both ESBL positive and negative isolates in a cluster highlights no correlation observed between the clusters' strains in the dendrogram on the basis of ESBL production. Cluster-A RAPD types indicate the possibility of the contamination of well water with clinical type strains that could be pathogenic for humans when ingested and can lead to diarrhea. DNA profiles generated enlighten that RAPD could be used as an efficient typing method for investigating outbreaks of significant nosocomial infections, discriminating the isolates, and reflecting the genotypes circulating either in the hospital setting or in the community. Future studies are required to establish an association between a particular RAPD type to the virulence of a pathogen in tracing its origin and source of infection, which will be helpful in molecular epidemiology studies and implementing control measures.

Our study has several limitations. Relatively low sample size and low ESBL-producing *K. pneumoniae* recovery rate limited our ability to elucidate clear genetic relatedness between healthcare and non-healthcare-associated isolates. Further, we did not screen the isolates for plasmid-mediated cephalosporinase and carbapenemase, although the isolates were phenotypically resistant to these class of antibiotics. Future studies with a larger number of samples from different environmental sources will provide more comprehensive information on enteric pathogens and β-lactam genes. The study was conducted at a single tertiary care hospital, and hence the results may not be generalized to other settings.

**Conclusion**

The high burden of *K. pneumoniae* carrying multiple drug resistance genes isolated from clinical and environmental samples from our study is alarming since it causes life-threatening infections. The findings draw attention to health practitioners in engaging in an effective antimicrobial stewardship program, judicious use of empirical antimicrobial therapy, emphasizing culture and molecular-based techniques for diagnosis wherever feasible and strict adherence to infection control policy.

Our findings also showed the majority of the clinical genotypes to be traced epidemiologically with well water, which was used as a drinking water source. This emphasizes the need for proper and frequent treatment of well water before consumption. Besides, there should be adequate knowledge provided to the fish vendors about the appropriate handling of fish and effective treatment of fish effluents before being let out into the environment, reducing the chances of infections among humans and the community. Further, public awareness towards excessive and non-therapeutic use of medically important antimicrobials will help achieve a one-health approach where different organizations can work towards the benefit of achieving better health outcomes.

**Acknowledgments**

This work was supported by NITTE (Deemed to be University) in the form of a Ph.D. scholarship and research funding support (grant no. NUFR3/ 2017/ 06/ 03). We gratefully acknowledge the Department of Microbiology, KS Hegde Medical Academy, NITTE (Deemed to be University) for providing facilities and the Dean and DBT Bioinformatics Centre, College of Fisheries, Mangaluru, for carrying the RAPD analysis work.

**Conflict of interest**: The authors declare no conflict of interest.

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Conceived and designed the experiments: SG, AVS, AKS. Performed the experiments: SG, PTG, MS. Analysed the data: SG, AVS, MS, AKS, PTG. Wrote the paper: SG. Manuscript revisions: AVS, SG, MS, AKS. All authors approved the final version of the manuscript and agreed to be held accountable for its content.

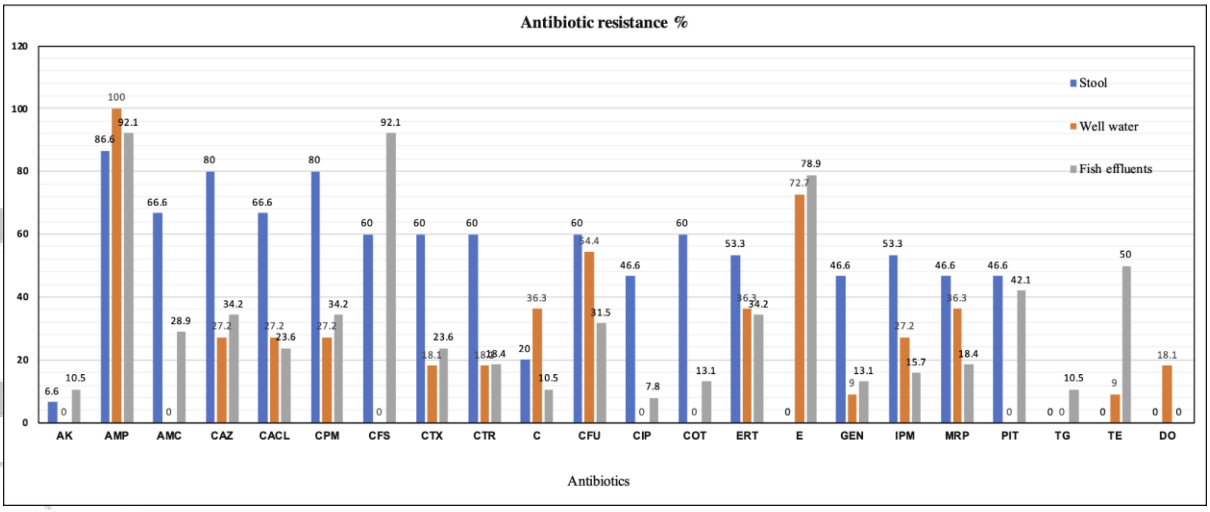


**Figure 1** Combined antibiotic resistance pattern of *K. pneumoniae* isolated from diarrheal stool, well water, and fish market effluent samples. (AK- amikacin; AMP- ampicillin; AMC- amoxyclav; CAZ- ceftazidime; CACL- ceftazidime clavulanic acid; CPM- cefepime; CFS- cefoperazone-sulbactam; CTX- cefotaxime; CTR- ceftriaxone; C- chloramphenicol; CFU- cefuroxime; CIP- ciprofloxacin; COT- cotrimoxazole; ERT- ertapenem; E- erythromycin; GEN- gentamicin; IPM- imipenem; MRP- meropenem; PIT- piperacillin tazobactam; TG- tigecycline; TE- tetracycline; DO- doxycycline)

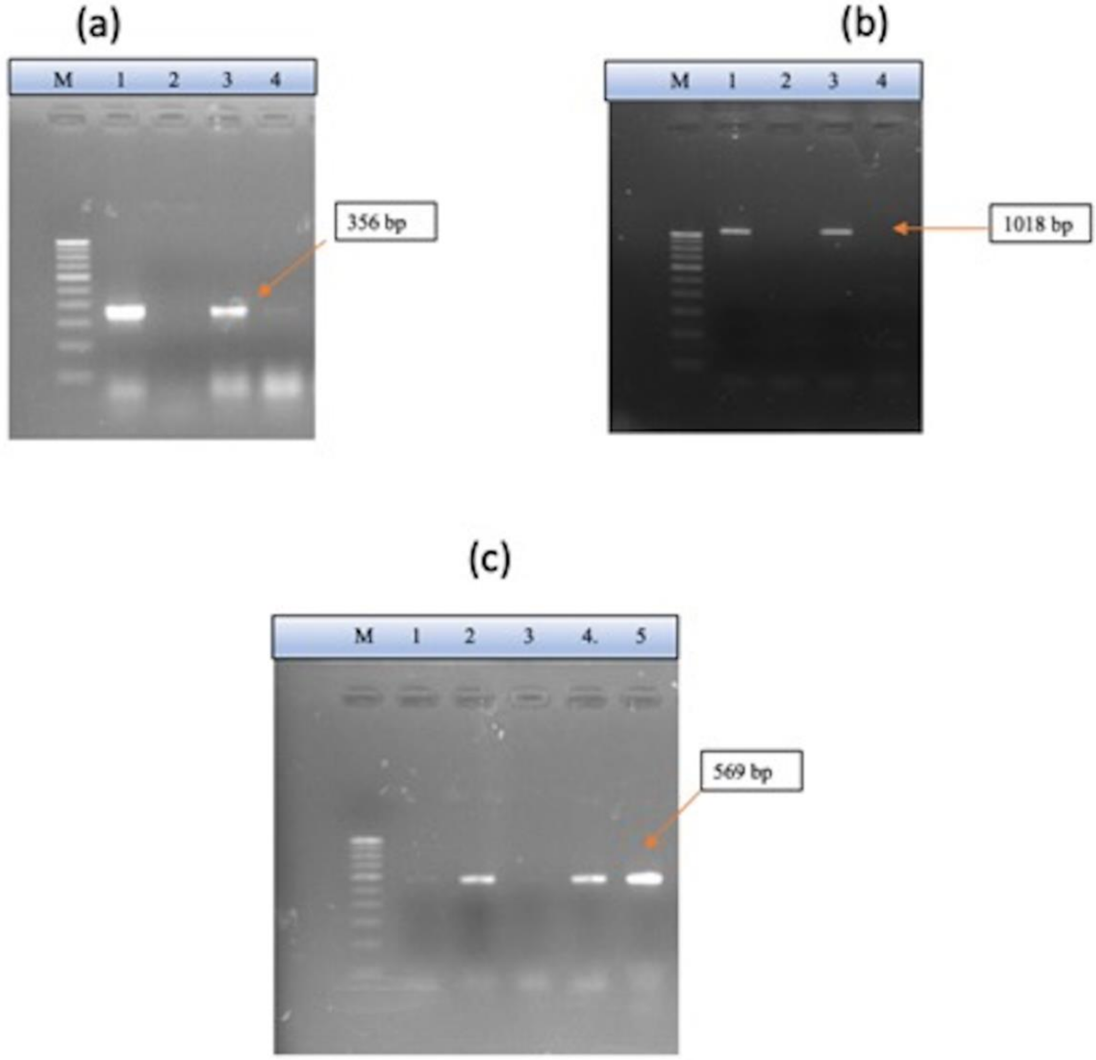


**Figure 2** Polymerase chain reaction images. (a) *bla*$_{\mathrm{CTX}}$ gene. Lane M: 100 bp DNA ladder; lane 1: positive control; lane 2: negative control; lanes 3 and 4: *K. pneumoniae* isolates. (b) *bla*$_{\mathrm{SHV}}$ gene. Lane M: 100 bp DNA ladder; lane 1: positive control; lane 2: negative control; lanes 3 and 4: *K. pneumoniae* isolates. (C) *bla*$_{\mathrm{TEM}}$ gene. Lane M: 100 bp DNA ladder; lane 1: negative control; lane 2: positive control; lanes 3,4 and 5: *K. pneumoniae* isolates

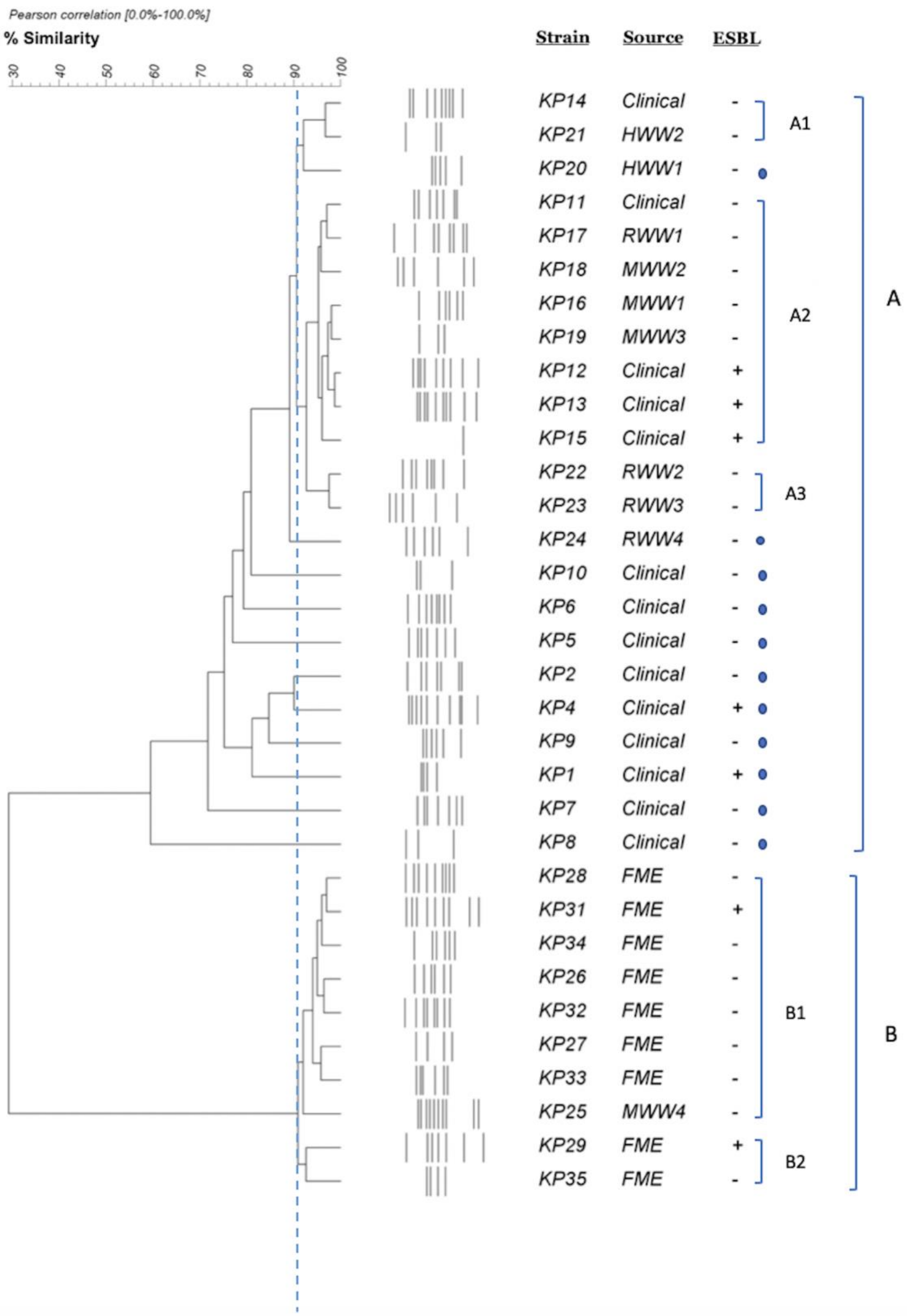


**Figure 3** Dendrogram representing random amplified polymorphic DNA profiles for *K. pneumoniae* isolates in this study. FME: Fish market effluent; HWW: Hospital well water, MWW: Market well water; RWW: Residential well water

**Table 1.** List of primers used in the study, their sequences and thermal cycling conditions.

| Target gene | Primer sequences | Thermal cycling conditions | Product size (bp) | Reference |
|---|---|---|---|---|
| *bla*$_{CTX}$ | F: ACGTTAAACACCGCCATTCC<br>R: TCGGTGACGATTTTAGCCGC | 95°C 5 min (1 cycle) 94°C 15 s, 60°C 1 min ,72°C 1.3 min (30 cycles), 72°C 4 min (1 cycle) | 356 | [Colomer-Lluch et al. 2011] |
| *bla*$_{SHV}$ | F: ATTTGTCGCTTCTTTACTCGC<br>R: TTTATGGCGTTACCTTTGACC | 94°C 5 min ,94°C 30 s (30 cycles), 52°C 30 s, 72° 50 s, 72°C 10 min | 1018 | [Jemima and Verghesese. 2008] |
| *bla*$_{TEM}$ | F: CTCACCCAGAAACGCTGGTG<br>R: ATCCGCCTCCATCCAGTCTA | 95°C 5 min (1 cycle), 94°C 15 s, 63°C 1 min, 72°C 1.3 min (30 cycles), 72°C 4 min (1 cycle) | 569 | [Colomer-Lluch et al. 2011] |
| Primer 640 | CGTGGGGCCT | 95°C 5 minutes, 35 cycles of 1 min at 94°C, 38°C for 2 min, 2 min at 72°C, 72°C for 10 min. | 10 | [Haryani et al. 2007] |
| Primer 650 | AGTATGCAGC | 95°C 5 minutes, 35 cycles of 1 min at 94°C, 38°C for 2 min, 2 min at 72°C, 72°C for 10 min. | 10 | [Ashayeri et al. 2012] |

**Table 2.** Antibiotic susceptibility profiles of *K. pneumoniae* isolates from clinical and environmental samples.

| Antibiotic class | Antibiotics | Fish market effluents (n=38) | | Stool samples (n=15) | | Well water (n=11) | | Fisher's exact value | *p*-Value |
|---|---|---|---|---|---|---|---|---|---|
| | | n (resistant %) | n (Susceptible %) | n (resistant %) | n (Susceptible %) | n (resistant %) | n (Susceptible %) | | |
| β–lactam antibiotics | Ampicillin | 35 (92.1%) | 3 (7.9%) | 13 (86.7%) | 2 (13.3%) | 11 (100.0%) | 0 (0.0%) | 1.306 | 0.672 |
| | Ceftazidime | 13 (34.2%) | 25 (65.8%) | 12 (80.0%) | 3 (20.0%) | 3 (27.3%) | 8 (72.7%) | 10.384 | 0.006* |
| | Cefepime | 13 (34.2%) | 25 (65.8%) | 12 (80.0%) | 3 (20.0%) | 3 (27.3%) | 8 (72.7%) | 10.384 | 0.006* |
| | Cefotaxime | 9 (23.7%) | 29 (76.3%) | 9 (60.0%) | 6 (40.0%) | 2 (18.2%) | 9 (81.8%) | 6.976 | 0.031* |
| | Ceftriaxone | 7 (18.4%) | 31 (81.6%) | 9 (60.0%) | 6 (40.0%) | 2 (18.2%) | 9 (81.8%) | 8.825 | 0.012* |
| | Cefuroxime | 12 (31.6%) | 26 (68.4%) | 9 (60.0%) | 6 (40.0%) | 6 (54.5%) | 5 (45.5%) | 4.376 | 0.125 |
| | Ertapenem | 13 (34.2%) | 25 (65.8%) | 8 (53.3%) | 7 (46.7%) | 4 (36.4%) | 7 (63.6%) | 1.714 | 0.426 |
| | Imipenem | 6 (15.8%) | 32 (84.2%) | 8 (53.3%) | 7 (46.7%) | 3 (27.3%) | 8 (72.7%) | 7.316 | 0.021* |
| | Meropenem | 7 (18.4%) | 31 (81.6%) | 7 (46.7%) | 8 (53.3%) | 4 (36.4%) | 7 (63.6%) | 4.739 | 0.093* |
| Non-β–Lactam antibiotics | Amikacin | 4 (10.5%) | 34 (89.5%) | 1 (6.7%) | 14 (93.3%) | 0 (0.0%) | 11 (100.0%) | 0.860 | 0.817 |
| | Chloramphenicol | 4 (10.5%) | 34 (89.5%) | 3 (20.0%) | 12 (80.0%) | 4 (36.4%) | 7 (63.6%) | 4.051 | 0.115 |
| | Ciprofloxacin | 3 (7.9%) | 35 (92.1%) | 7 (46.7%) | 8 (53.3%) | 0 (0.0%) | 11 (100.0%) | 11.660 | 0.002* |
| | Cotrimoxazole | 5 (13.2%) | 33 (86.8%) | 9 (60.0%) | 6 (40.0%) | 0 (0.0%) | 11 (100.0%) | 14.983 | 0.001* |
| | Erythromycin | 30 (78.9%) | 8 (21.1%) | 0 (0.0%) | 0 (0.0%) | 8 (72.7%) | 3 (27.3%) | 4.686 | 0.077* |
| | Gentamicin | 5 (13.2%) | 33 (86.8%) | 7 (46.7%) | 8 (53.3%) | 1 (9.1%) | 10 (90.9%) | 7.207 | 0.020* |
| β–Lactam inhibitors | Ampicillin clavulanic acid | 11 (28.9%) | 27 (71.1%) | 10 (66.7%) | 5 (33.3%) | 0 (0.0%) | 11 (100.0%) | 13.442 | 0.001* |
| | Ceftazidime clavulanic acid | 9 (23.7%) | 29 (76.3%) | 10 (66.7%) | 5 (33.3%) | 3 (27.3%) | 8 (72.7%) | 8.523 | 0.012* |
| | Cefoperazone sulbactam | 35 (92.1%) | 3 (7.9%) | 9 (60.0%) | 6 (40.0%) | 0 (0.0%) | 11 (100.0%) | 34.729 | 0.001* |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Piperacillin tazobactam | 30 (78.9%) | 8 (21.1%) | 7 (46.7%) | 8 (53.3%) | 0 (0.0%) | 11 (100.0%) | 24.172 | 0.001* |

* significant *p*- value < 0.05

**Table 3.** Antibiotic profile and Multiple antibiotic resistance (MAR) indices of *K. pneumoniae* isolates.

| Strain No. | Source | Antibiotic resistance profile | MAR index |
|---|---|---|---|
| KP-1 | Clinical | GEN | 0.05 |
| KP-2 | Clinical | AMC, GEN | 0.1 |
| KP-3 | Clinical | AMP, CA, CPM, CFU, CTR, CTX , ERT, IMP | 0.38 |
| KP-4 | Clinical | AMC, AMP, CA, CACL, COT, CPM, | 0.29 |
| KP-5 | Clinical | AMC, AMP, CA, CACL, CFS, CFU, CIP, CPM, COT, CTR , CTX, ERT, GEN, IMP, MRP, PT | 0.76 |
| KP-6 | Clinical | AMC, AMP, CA, CACL, CIP, COT,CPM, CTR, CFU, CTX, ERT, IMP, MRP | 0.62 |
| KP-7 | Clinical | AMP, COT | 0.1 |
| KP-8 | Clinical | AMC, AMP, CA, CACL, CFS, COT, CPM | 0.33 |
| KP-9 | Clinical | AMC, AMP, AK, CA, CACL, CFS, CFU, CIP, CPM, CTR, CTX, ERT, IMP, MRP, PT | 0.71 |
| KP-10 | Clinical | AMP, C, CA, CFS, CPM, CFU, CTR, CTX, ERT | 0.43 |
| KP-11 | Clinical | AMP, CA, CACL, CFS, CPM, GEN, PT | 0.33 |
| KP-12 | Clinical | AMC, AMP, CA, CACL, CFS, CIP, COT, CPM, CTR, CTX, CFU, GEN, IMP, MRP, PT | 0.71 |
| KP-13 | Clinical | AMC, AMP, CA, CACL, CFS, CFU, CIP, COT, CPM, CTR, CTX, ERT, IMP, MRP, PT | 0.71 |
| KP-14 | Clinical | AMC, AMP, C, CA, CACL, CFS, CFU, CIP,COT, CPM, CTR, CTX, ERT, GEN, IMP, MRP, PT | 0.81 |
| KP-15 | Clinical | AMC, AMP, C, CA, CACL, CFS, CFU, CIP, COT, CPM, CTR, CTX, ERT, GEN, IMP, MRP, PT | 0.81 |
| KP-16 | Market Well water | AMP,CFU,E, ERT | 0.19 |
| KP-17 | Residential Well water | AMP, C, ERT, MRP | 0.19 |
| KP-18 | Market Well water | AMP, E, IMP | 0.14 |
| KP-19 | Market Well water | AMP, C, CA, CACL, CFU, CPM, CTR, CTX | 0.38 |
| KP-20 | Hospital Well water | AMP, CFU, DOXY, E, MRP, TETRA | 0.29 |
| KP-21 | Hospital Well water | AMP, C, CFU, E | 0.19 |
| KP-22 | Residential Well water | AMP, E, ERT | 0.14 |
| KP-23 | Residential Well water | AMP, CFU, DOXY, E, GEN, IMP, MRP | 0.33 |
| KP-24 | Residential Well water | AMP, CA, CACL, CPM, E | 0.24 |
| KP-25 | Market Well water | AMP, C, CA , CACL, CFU, CPM, CTR, CTX, ERT, IMP, MRP | 0.52 |
| KP-26 | Fish Market Effluent | AK, AMP, CA, CACL, CFS, CPM, CTX | 0.33 |
| KP-27 | Fish Market Effluent | AMP, CA, CACL, CFS, CTR, CTX, E, MRP, PT | 0.43 |
| KP-28 | Fish Market Effluent | AMP, CA, CACL, CFS, COT, CPM, E, PT, TG | 0.43 |
| KP-29 | Fish Market Effluent | AMP, CA, CACL, CFS, CFU, CTX, E, PT | 0.38 |
| KP-30 | Fish Market Effluent | AMP, CA, CACL, CFS, E, PT | 0.29 |
| KP-31 | Fish Market Effluent | AMP, CFS, E, PT | 0.19 |
| KP-32 | Fish Market Effluent | C, CFS, E, MRP, PT | 0.24 |
| KP-33 | Fish Market Effluent | AK, AMP, CFS, E, PT | 0.24 |
| KP-34 | Fish Market Effluent | AMC, AMP, C, CA, CACL, CFS, CPM, E, ERT, MRP, PT, TETRA | 0.57 |
| KP-35 | Fish Market Effluent | AMC, AMP, C, CA, CFS, CFU, COT, CPM, E, ERT, IMP, MRP, PT, TETRA, TG | 0.71 |

**Table 4.** Dendrogram-cluster-based antibiotic resistance pattern of *K. pneumoniae* isolates.

| Clusters | Strain No. | Sample type | Antibiotic resistance phenotypes |
|---|---|---|---|
| | | **Cluster- A (Three clusters)** | |
| **Cluster- A1** | KP-14 | Clinical | AMC, AMP, C, CA,CACL, CFS, CFU, CIP,COT, CPM, CTR, CTX, ERT, GEN, IMP, MRP, PT |
| | KP-21 | Hospital Well water | AMP, C, CFU, E |
| **Cluster- A2** | KP-11 | Clinical | AMP, CA, CACL, CFS, CPM, GEN, PT |
| | KP-12 | Clinical | AMC, AMP, CA, CACL, CFS, CIP, COT, CPM, CTR, CTX, CFU, GEN, IMP, MRP, PT |
| | KP-13 | Clinical | AMC, AMP, CA, CACL, CFS, CFU, CIP, COT, CPM, CTR, CTX, ERT, IMP, MRP, PT |
| | KP-7 | Clinical | AMP, COT |
| | KP-15 | Clinical | AMC, AMP, C, CA, CACL, CFS, CFU, CIP, COT, CPM, CTR, CTX, ERT, GEN, IMP,MRP, PT |
| | KP-16 | Market Well water | AMP,CFU,E, ERT |
| | KP-17 | Residential Well water | AMP, C, ERT, MRP |
| | KP-18 | Market Well water | AMP, E, IMP |
| | KP-19 | Market Well water | AMP, C, CA, CACL, CFU, CPM, CTR, CTX |
| **Cluster- A3** | KP-22 | Residential Well water | AMP, E, ERT |
| | KP-23 | Residential Well water | AMP, CFU, DOXY, E, GEN, IMP, MRP |
| | | **Cluster- B (Two clusters)** | |
| **Cluster- B1** | KP-25 | Market Well water | AMP, C, CA , CACL, CFU, CPM, CTR, CTX, ERT, IMP, MRP |
| | KP-26 | Fish Market Effluent | AK, AMP, CA, CACL, CFS, CPM, CTX |
| | KP-27 | Fish Market Effluent | AMP, CA, CACL, CFS, CTR, CTX, E, MRP, PT |
| | KP-28 | Fish Market Effluent | AMP, CA, CACL, CFS, COT, CPM, E, PT, TG |
| | KP-31 | Fish Market Effluent | AMP, CFS, E, PT |
| | KP-32 | Fish Market Effluent | C, CFS, E, MRP, PT |
| | KP-33 | Fish Market Effluent | AK, AMP, CFS, E, PT |
| | KP-34 | Fish Market Effluent | AMC, AMP, C, CA, CACL, CFS, CPM, E, ERT, MRP, PT, TETRA |
| **Cluster- B2** | KP-29 | Fish Market Effluent | AMP, CA, CACL, CFS, CFU, CTX, E, PT |
| | KP-35 | Fish Market Effluent | AMC, AMP, C, CA, CFS, CFU, COT, CPM, E, ERT, IMP, MRP, PT, TETRA, TG |

**Practitioner points:**

- Detection of multi-drug resistant *Klebsiella* strains in hospital wastewater and drinking water sources has progressively increased since its emerging resistance to third-generation cephalosporins and carbapenems.
- Detection of *beta-lactamase encoding* genes by molecular techniques and typing by random amplified polymorphic DNA (RAPD) can be useful in identifying the genetic fingerprints for epidemiological study.
- Implementation of effective antimicrobial stewardship program and infection control policy thereby helps assess the risk factors associated with infections.

Graphical Abstract